

$^{90}\text{Y}$   $\beta^-$  decay (3.19 h) 1976Gr16

| Type            | Author                      | History | Citation         | Literature Cutoff Date |
|-----------------|-----------------------------|---------|------------------|------------------------|
| Full Evaluation | S. K. Basu, E. A. Mccutchan |         | NDS 165,1 (2020) | 1-Mar-2020             |

Parent:  $^{90}\text{Y}$ : E=682.01 5;  $J^\pi=7^+$ ;  $T_{1/2}=3.19$  h 1;  $Q(\beta^-)=2278.5$  16;  $\% \beta^-$  decay=0.0018 2

$^{90}\text{Y}$ - $\% \beta^-$  decay: From measured  $I(2319\gamma)/I(480\gamma)=1.89 \times 10^{-5}$  18 (1976Gr16); in agreement with upper limit of  $8 \times 10^{-5}$  (1973Ra10).

Value is recalculated by evaluator with adopted photon branching ratios.

1976Gr16: Y(n, $\gamma$ ), chemical separation. Measured  $\gamma(t)$  with Ge(Li).

Other: 1973Ra10.

$\alpha$ : Additional information 1.

 $^{90}\text{Zr}$  Levels

| E(level) <sup>†</sup> | $J^\pi$ | $T_{1/2}$ <sup>†</sup> |
|-----------------------|---------|------------------------|
| 0                     | $0^+$   | stable                 |
| 2319.000 9            | $5^-$   | 809.2 ms 20            |

<sup>†</sup> From the Adopted Levels.

 $\beta^-$  radiations

| E(decay)   | E(level) | $I\beta^-$ <sup>†</sup> | Log $ft$             | Comments                |
|------------|----------|-------------------------|----------------------|-------------------------|
| (641.5 16) | 2319.000 | 0.0018 2                | 9.62 <sup>1u</sup> 5 | av E $\beta$ =231.33 62 |

<sup>†</sup> Absolute intensity per 100 decays.

 $\gamma(^{90}\text{Zr})$ 

| $E_\gamma$ <sup>†</sup> | $I_\gamma$ <sup>‡</sup> | $E_i(\text{level})$ | $J_i^\pi$ | $E_f$ | $J_f^\pi$ | Mult. <sup>†</sup> | $\alpha$              | Comments                                                                                                                                                               |
|-------------------------|-------------------------|---------------------|-----------|-------|-----------|--------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2318.968 10             | 100                     | 2319.000            | $5^-$     | 0     | $0^+$     | E5                 | $4.64 \times 10^{-4}$ | $\alpha(K)=0.000408$ 6; $\alpha(L)=4.63 \times 10^{-5}$ 7; $\alpha(M)=8.04 \times 10^{-6}$ 12; $\alpha(N)=1.141 \times 10^{-6}$ 16; $\alpha(O)=7.97 \times 10^{-8}$ 12 |

<sup>†</sup> From the Adopted Gammas.

<sup>‡</sup> For absolute intensity per 100 decays, multiply by 0.000018 2.

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Decay Scheme

Intensities:  $I_{(\gamma+ce)}$  per 100 parent decays

